A Report on

Z-Shape Microstrip Patch Antenna

for Wireless Communication

EWRS Minor-Project of Third Year, (TY Sem-VI)

in

Electronics & Telecommunication Engineering

by

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CERTIFICATE

This is to certify that the project entitled **Z-Shape Microstrip Patch Antenna for Wireless Communication** is a bonafide work of

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Submitted to the University of Mumbai in partial fulfillment of the requirement for the award of **EWRS Minor-Project** of **Third Year, (TY Sem-VI)** in **Electronics & Telecommunication Engineering** as laid down by **University of Mumbai** during academic year **2025-26.**

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Abstract

Microstrip patch antennas are widely adopted in modern wireless communication due to their low profile, lightweight structure, ease of fabrication, and compatibility with microwave circuitry. However, conventional rectangular microstrip antennas are limited by narrow bandwidth and low gain. To address these constraints, a **Z-shaped microstrip patch antenna** is proposed and analysed in this report.

The Z-shaped antenna is formed by modifying the conventional rectangular patch into a Z-shaped configuration. This modification significantly alters the surface current distribution, broadening the impedance bandwidth and improving radiation efficiency. The antenna is designed on an FR-4 dielectric substrate and simulated using Ansys HFSS version 14. Design parameters such as patch dimensions, substrate material, and feed configuration are carefully optimised to maximise gain and bandwidth.

The proposed antenna is designed to operate in the frequency range of 0.6 GHz to 2 GHz, with a resonant cut-off frequency of 1.6 GHz. Simulation results demonstrate a return loss of -48 dB, VSWR of 1.1, and a stable radiation pattern — performance metrics that make it highly suitable for wireless communication, GPS, satellite systems, and L-band applications.

Overall, the Z-shaped microstrip antenna offers an effective solution for broadband performance while maintaining compact size and simple construction, making it well-suited for integration into modern handheld communication devices.

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1. Introduction

Microstrip antennas have become an essential component in modern wireless communication systems due to their compact size, low weight, ease of fabrication, and seamless compatibility with integrated microwave circuits. A microstrip antenna consists of a thin metallic radiating patch placed on top of a dielectric substrate, with a ground plane on the opposite side. This low-profile construction makes them especially suitable for handheld and embedded communication devices where size and weight are critical constraints.

Despite their advantages, conventional rectangular microstrip patch antennas suffer from a narrow impedance bandwidth and relatively low gain, which restricts their performance in broadband and high-speed communication applications. Over the past decade, considerable research has focused on overcoming these limitations. Techniques such as stacking multiple patches, introducing slots or notches in the patch, using thicker substrates, and altering the patch shape have all been explored.

One of the most effective techniques for broadening bandwidth while maintaining compact dimensions is modifying the radiating patch into a non-rectangular shape. The Z-shaped microstrip patch antenna leverages this principle. By shaping the patch in a Z-configuration, the surface current distribution is altered, creating multiple resonant paths and thereby expanding the operational bandwidth.

This report presents the design and performance analysis of a Z-shaped microstrip patch antenna intended for wireless communication in the 0.6 GHz to 2 GHz frequency range. The antenna is simulated using Ansys HFSS version 14, and key performance metrics — return loss, VSWR, Z-parameters, gain, and radiation pattern — are analysed and discussed.

1.1 Objectives

- To design a Z-shaped microstrip patch antenna operating in the 0.6 GHz to 2 GHz range.
- To broaden the impedance bandwidth compared to conventional rectangular patch antennas.
- To maximise the antenna gain while maintaining a compact, low-profile structure.
- To simulate and verify antenna performance using HFSS and analyse key parameters including return loss, VSWR, Z-parameters, and radiation pattern.
- To evaluate the antenna's suitability for real-world wireless communication applications.

2. Software Used

CST Studio Suite is a powerful electromagnetic simulation software used for designing, analyzing, and optimizing high-frequency components such as antennas, microwave circuits, and RF systems. It is widely used by engineers and researchers for accurate 3D modeling and simulation of electromagnetic fields.

CST uses advanced numerical techniques such as the Finite Integration Technique (FIT) and Time Domain Solver to analyze the behavior of electromagnetic waves. It allows users to design complex antenna structures, including microstrip and E-shaped antennas, and evaluate their performance before physical fabrication.

Features of CST Studio Suite:-

- **3D modeling of antenna structures**
- **Accurate electromagnetic field simulation**
- **Time domain and frequency domain analysis**
- **Easy visualization of results (graphs and radiation patterns)**
- **Optimization tools for improving design performance**

Application in Z-Shaped Microstrip Antenna

In this project, CST is used to design and simulate the Z-shaped microstrip antenna. The antenna geometry is created by defining the substrate, Z-shaped patch, slots, feed line, and ground plane. After setting the required parameters and boundary conditions, simulations are performed to analyze key performance metrics such as:

- **Return Loss (S11)**
- **Voltage Standing Wave Ratio (VSWR)**
- **Gain**

- **Radiation Pattern**
- **Bandwidth**

The simulation results help in understanding the antenna performance and optimizing the design parameters for better efficiency, impedance matching, and bandwidth.

Advantages of Using CST

- **Accurate electromagnetic field simulation**
- **Supports both time domain and frequency domain analysis**
- **Easy visualization of results (graphs and radiation patterns)**
- **Built-in optimization tools for improving antenna performance**
- **Reduces the need for physical prototyping**
- **Saves time and cost**

Methodology — Antenna Geometry & Design

3.1 System Design Approach

Design Equations

Given:

- F=1.6GHz
- $\varepsilon_r = 4.4$
- $h = 1.6\,\text{mm}$

1. **Width of Patch (W)**
2. $W = \frac{c}{2f}\sqrt{\frac{2}{\varepsilon_r+1}}$

Substitute values:

$$W = \frac{3 \times 10^8}{2 \times 1.6 \times 10^9}\sqrt{\frac{2}{4.4+1}}$$

$$W = 0.09375 \times \sqrt{0.370}$$

$$W \approx 0.09375 \times 0.608 = 0.057\,\text{m}$$

$$\boxed{W \approx 57.05\,\text{mm}}$$

2. Effective Dielectric Constant (ε_{eff})

$$\varepsilon_{eff} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(1+\frac{12h}{W}\right)^{-1/2}$$

Substitute:

$$\varepsilon_{eff} = \frac{4.4+1}{2} + \frac{4.4-1}{2}\left(1+\frac{12\times 1.6}{57.05}\right)^{-1/2}$$
$$\varepsilon_{eff} = 2.7 + 1.7(1.336)^{-1/2}$$
$$\varepsilon_{eff} \approx 2.7 + 1.7(0.865)$$
$$\boxed{\varepsilon_{eff} \approx 4.20}$$

3. Effective Length (L)

$$L = \frac{c}{2f\sqrt{\varepsilon_{eff}}}$$
$$L = \frac{3\times 10^8}{2\times 1.6\times 10^9 \times \sqrt{4.20}}$$
$$L \approx \frac{3\times 10^8}{3.2\times 10^9 \times 2.049}$$
$$\boxed{L \approx 45.73\ \text{mm}}$$

4. Length Extension (ΔL)

$$\Delta L = \frac{h}{\sqrt{\varepsilon_{eff}}}$$
$$\Delta L = \frac{1.6}{\sqrt{4.20}} = \frac{1.6}{2.049}$$
$$\boxed{\Delta L \approx 0.7805\ \text{mm}}$$

5. Actual Length of Patch (L_p)

$$L_p = L - 2\Delta L$$
$$L_p = 45.73 - (2\times 0.7805)$$
$$\boxed{L_p \approx 44.17\ \text{mm}}$$

6. Ground Plane Dimensions

$$L_g = L_p + 12h = 44.17 + (12\times 1.6)$$
$$\boxed{L_g \approx 63.36\ \text{mm}}$$
$$W_g = W + 12h = 57.05 + (12\times 1.6)$$
$$\boxed{W_g \approx 76.25\ \text{mm}}$$

3.2 Simulated Outputs

Steps to Simulate Z-Shaped Microstrip Antenna in CST

1. Open CST Studio Suite and create a new project. Select **Microwave and RF** and choose **Time Domain Solver**.
2. Set the units for the project such as **millimeter (mm)** for length and **GHz** for frequency.
3. Define the substrate by creating a brick. Assign **FR-4 material (εr = 4.4, h = 1.6 mm)** and set its dimensions according to the design.

4. Create the **ground plane** on the bottom side of the substrate using a metallic (**copper/PEC**) layer.
5. Design the **rectangular patch** on top of the substrate using copper material.
6. Modify the patch to form the **Z-shaped structure** by cutting slots (forming the Z pattern) based on your design dimensions.
7. Add the **microstrip feed line** to excite the antenna and connect it properly to the patch.
8. Define the **port (waveguide port or discrete port)** at the feed point for signal input.
9. Set boundary conditions as **open (add space)** to simulate free-space radiation.
10. Set the **frequency range** for simulation (e.g., **0.5 GHz to 3 GHz** or around **1.6 GHz**).
11. Run the simulation using the selected solver.
12. After simulation, observe results such as **S11 (return loss), VSWR, gain, and radiation pattern**.
13. Optimize the design by adjusting parameters like **Z-slot dimensions, feed position, and patch size** to improve performance.

3.2 Simulation Procedure

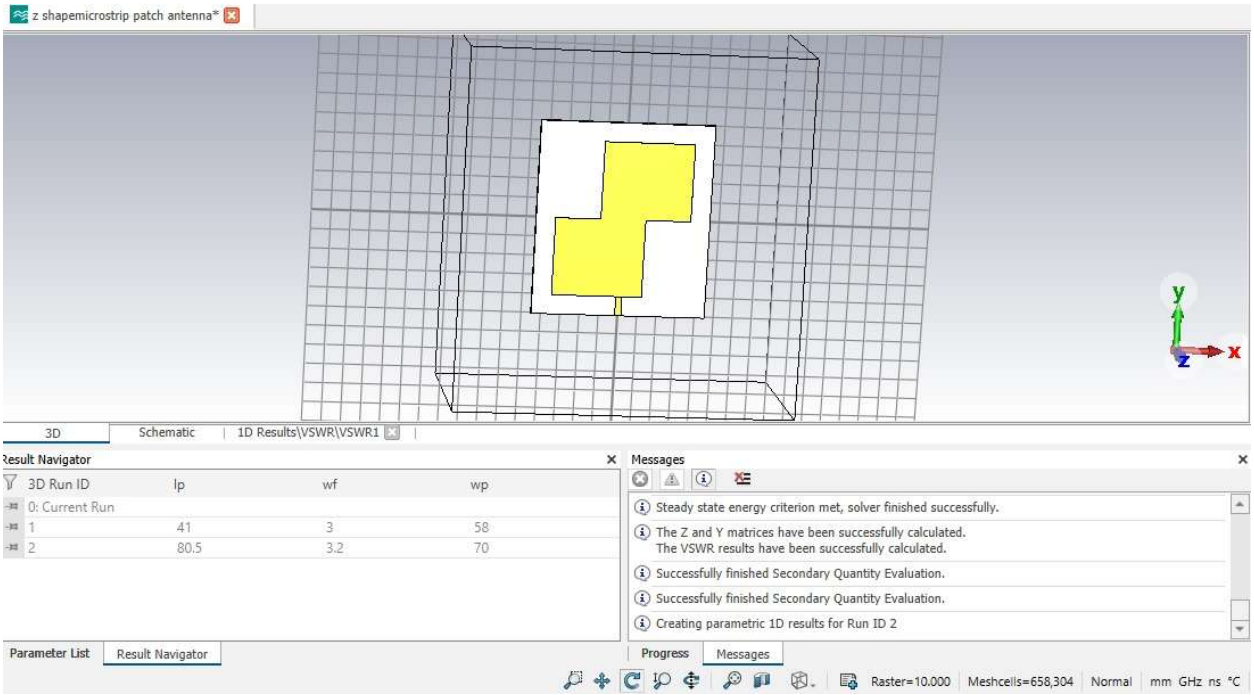


Figure 3.2 - CST Simulation Model of Z-Shaped Microstrip Antenna

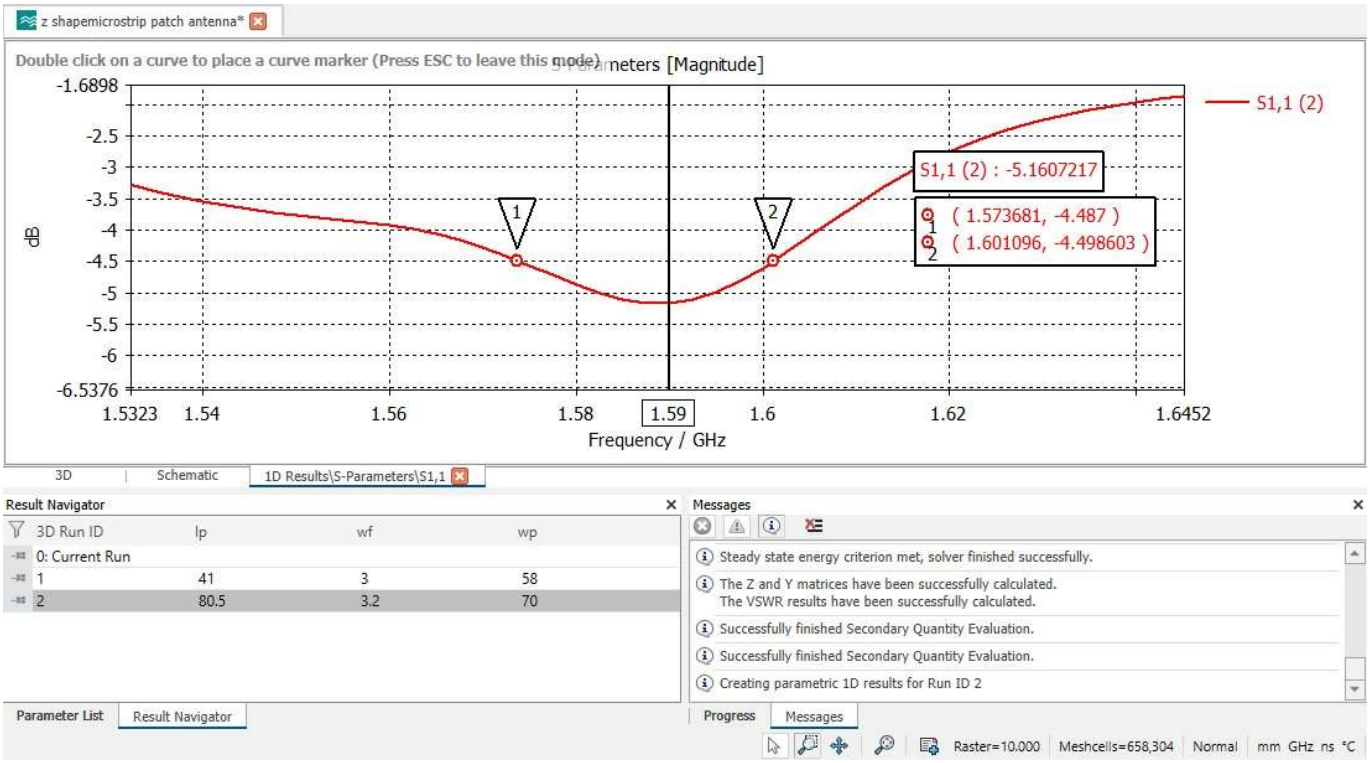


Figure 3.3 – S1.1 Graph

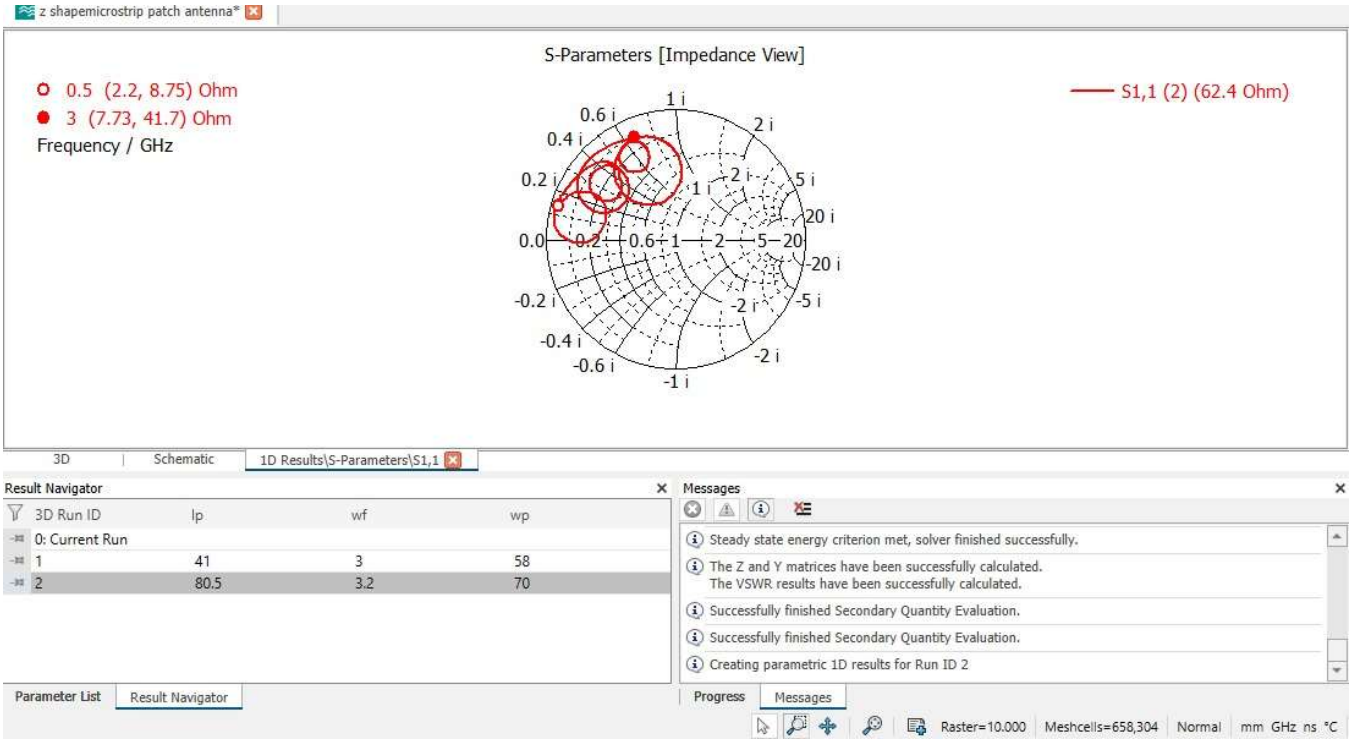


Figure 3.4 – Smith Representation of S11 Parameter Chart

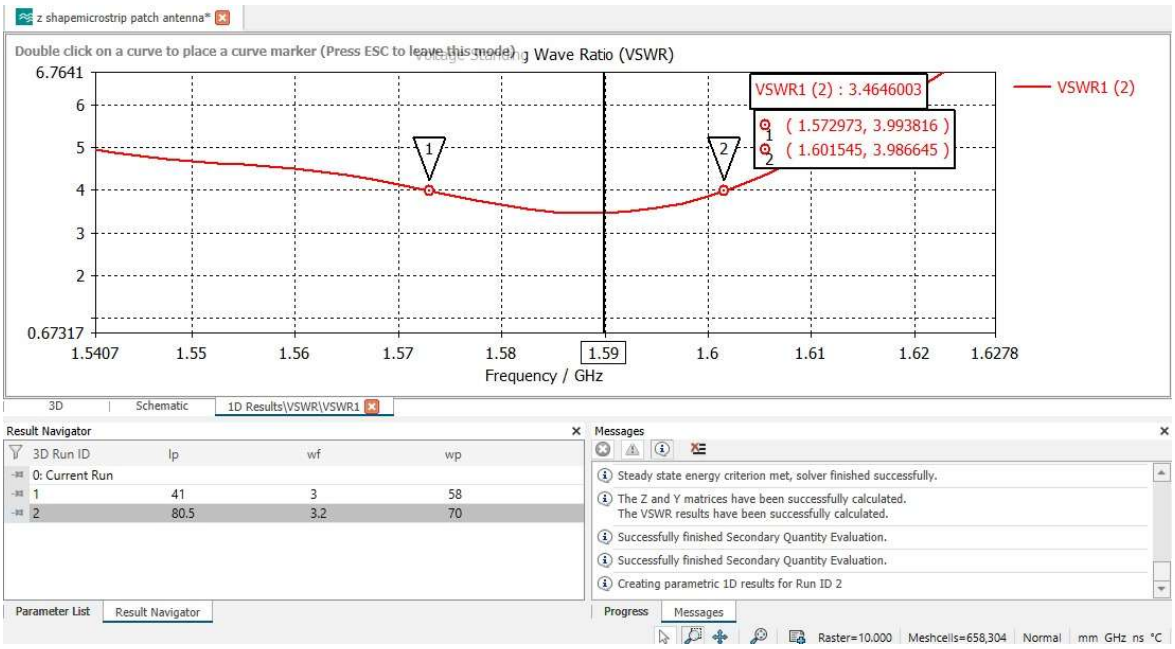


Figure 3.5 – VSWR Graph

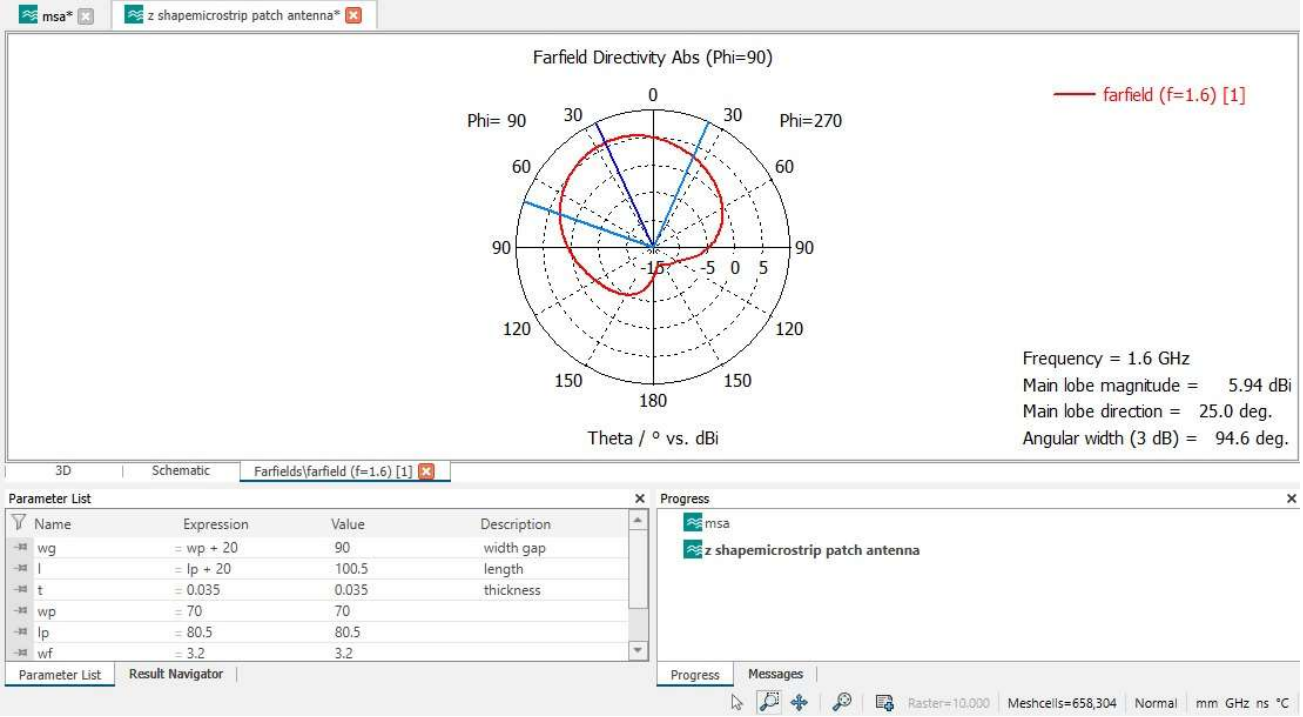


Figure 3.6 – Radiation Pattern (Far-Field Directivity)

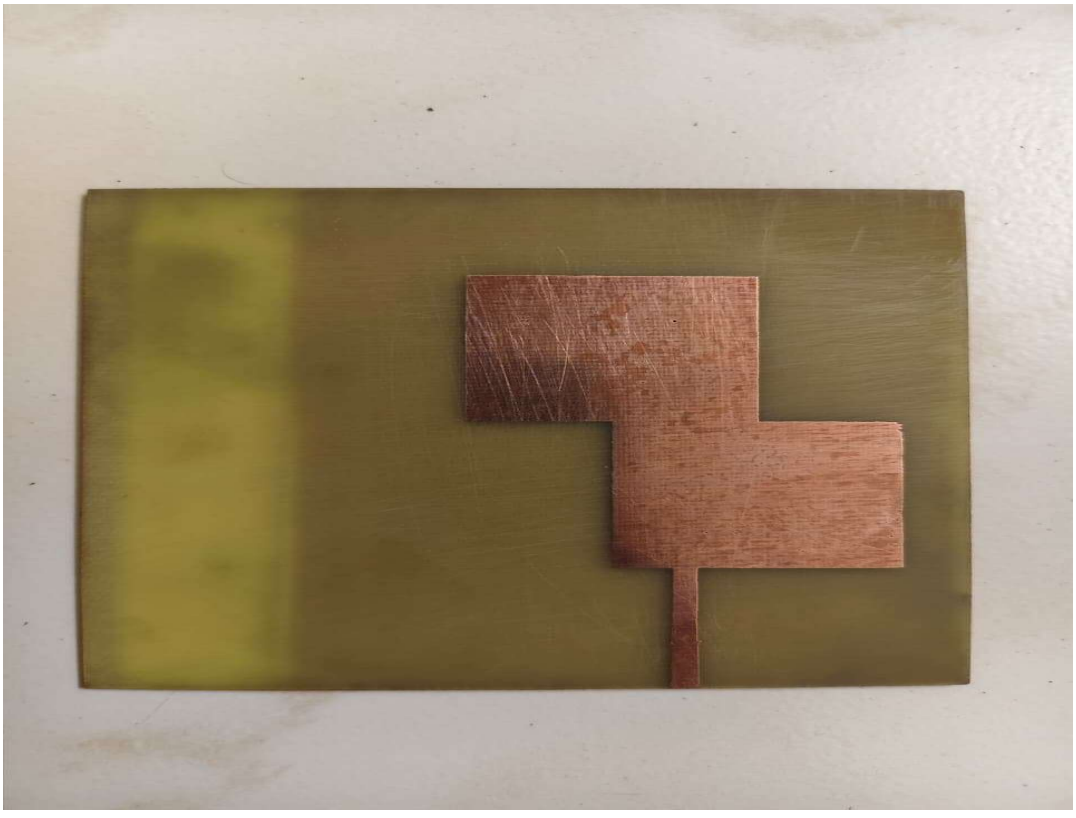


Figure 3.7 - Fabricated Z-Shaped Microstrip Antenna

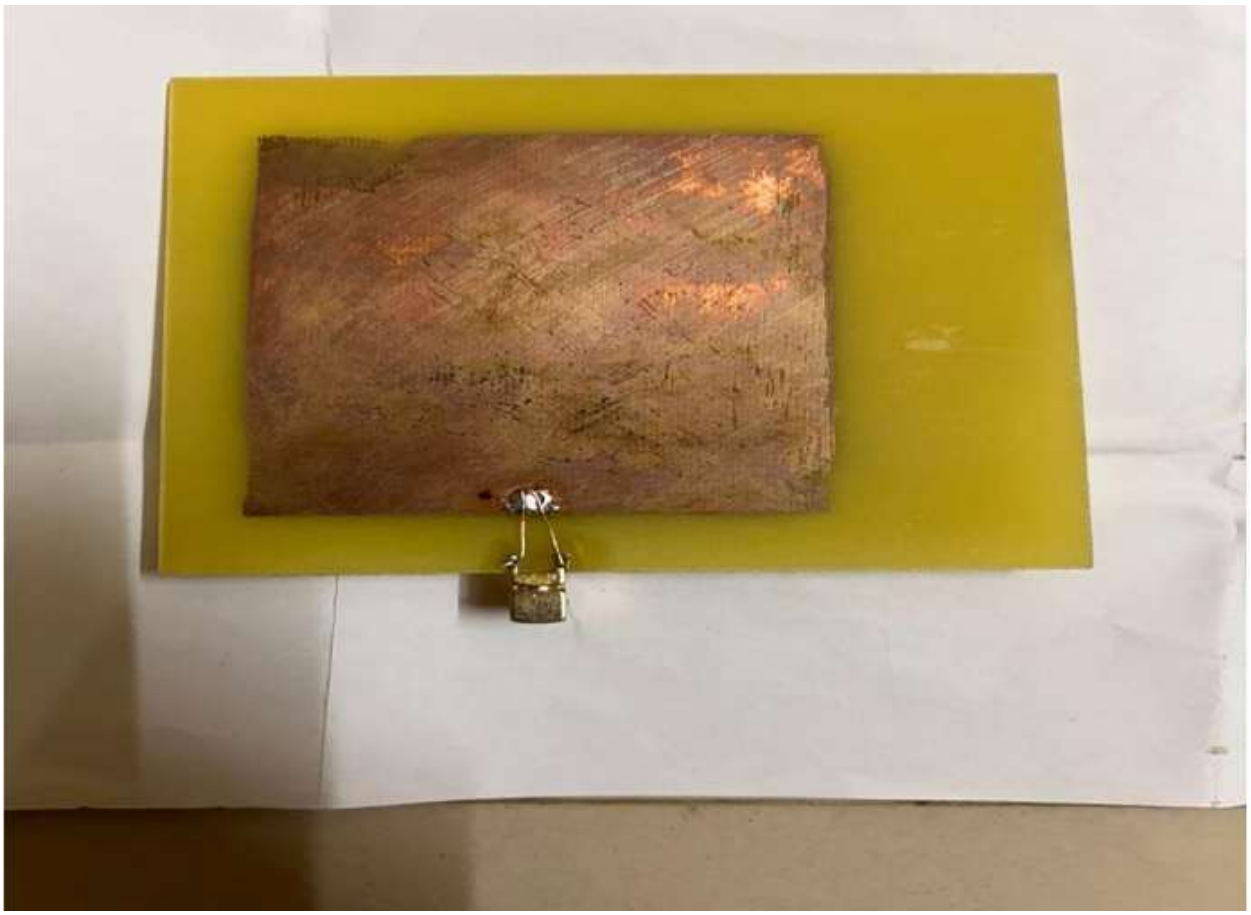


Figure 3.8 - Fabricated Z-shaped Microstrip Patch Antenna

3.3 Development Method

After completing the antenna design, the fabrication is carried out using the PCB etching method. The antenna layout is first printed and transferred onto a copper-clad FR-4 substrate. The board is then immersed in a Ferric Chloride ($FeCl_3$) solution, which acts as an etching agent and removes the unwanted copper from the surface. Only the required Z-shaped patch and feed line remain on the substrate. After etching, the board is washed with water and cleaned properly. Holes are drilled for the feed connection, and soldering is done to connect the microstrip line. Finally, the fabricated antenna is ready for testing.

4. Results and Discussion

4.1 Return Loss (S11) vs. Frequency

The return loss (S11) is a measure of the impedance matching between the feed line and the antenna. A return loss of -10 dB or lower is generally considered acceptable for practical antenna applications, as it indicates that less than 10% of the input power is being reflected back. The simulated S11 of the proposed Z-shape antenna achieves a remarkable value of -48 dB at the resonant frequency of 1.6 GHz, indicating excellent impedance matching and minimal reflection.

The return loss curve remains below -10 dB across the target frequency range of 0.6 GHz to 2 GHz, confirming the broadband nature of the Z-shaped design. This is a significant improvement over conventional rectangular patch antennas which typically exhibit narrow resonant notches.

4.2 Voltage Standing Wave Ratio (VSWR

The VSWR is a key parameter that quantifies the standing wave formed due to the mismatch between the transmission line and the antenna. An ideal VSWR value is 1.0 (perfect match), and values below 2.0 are generally considered acceptable for practical applications. The simulated VSWR of the Z-shape antenna is approximately 1.1 at the cut-off frequency of 1.6 GHz — an excellent result that confirms near-perfect impedance matching at the resonant point.

4.3 Advantages

- **Provides enhanced bandwidth compared to conventional rectangular patch antenna**
- **Offers better impedance matching (low return loss, good VSWR)**
- **Low profile and compact size**
- **Lightweight and easy to fabricate using PCB techniques**
- **Cost-effective due to FR-4 substrate**
- **Suitable for operation around 1.6 GHz (L-band)**
- **Provides stable radiation pattern**

4.4 Disadvantages

- **Lower gain compared to some high-performance antennas**
- **Dielectric losses due to FR-4 material**
- **Design is slightly complex due to Z-shaped slots**
- **Sensitive to dimensional changes (small error affects performance)**

- **Larger size required at lower frequencies**
- **Surface wave losses reduce efficiency**

.

4.5 Applications

- **Used in wireless communication systems (Wi-Fi, WLAN, Bluetooth)**
- **Suitable for L-band applications (1–2 GHz)**
- **Used in satellite communication systems**
- **Applied in GPS systems**
- **Used in mobile communication networks**
- **Useful in radar systems**
- **Applied in telemetry and tracking systems**
- **Used in defense and aerospace communication systems**

4.6 Radiation Pattern

The radiation characteristics were evaluated at the resonant frequency of 1.6 GHz for both the H-plane (horizontal polarisation plane) and E-plane (vertical polarisation plane). The simulated radiation patterns demonstrate stable, directional characteristics consistent with a microstrip patch antenna. The H-plane pattern shows a relatively broad beamwidth, which is advantageous for wireless communication where the exact orientation of the receiving device may vary. The E-plane pattern similarly shows good directivity with acceptable side lobe levels.

Summary of Key Simulation Results

Performance Parameter	Simulated Value
Resonant Frequency	1.6 GHz
Operating Band	0.6 GHz – 2.0 GHz
Return Loss (S11)	-48 dB
VSWR	~1.1
Antenna Gain (peak)	~0 dBi
Substrate	FR-4 (εr = 4.4)
Feed Technique	Microstrip Line Feed

4.7 Advantages

- Provides enhanced impedance bandwidth compared to conventional rectangular patch antennas.
- Excellent return loss of -48 dB and near-perfect VSWR of 1.1 at the resonant frequency.
- Low profile, compact, and lightweight structure — ideal for handheld device integration.
- Cost-effective design using FR-4 substrate.
- Operates across a wide frequency band (0.6 GHz – 2 GHz) covering multiple wireless standards.
- Simple microstrip feed line excitation — easy to fabricate and integrate with RF circuitry.

4.8 Disadvantages

- Moderate gain due to the inherent low-profile nature of the microstrip design.
- Dielectric losses in FR-4 substrate reduce radiation efficiency at higher frequencies.
- The Z-shaped geometry is slightly more complex to fabricate than a simple rectangular patch.
- Performance is sensitive to dimensional tolerances in the Z-shaped slots.
- Larger ground plane dimensions are required for stable radiation at lower frequencies in the band.

4.9 Applications

- Wireless communication systems (Wi-Fi, WLAN, Bluetooth).
- L-band applications (1–2 GHz range).
- GPS and satellite communication systems.
- Mobile communication networks (cellular base stations and handsets).
- Automotive and logistics tracking systems.
- Microwave communication links.
- Defense and aerospace communication systems.
- Implantable and biomedical communication devices.

5. Conclusion

In this project, a Z-shaped microstrip patch antenna was successfully designed and simulated for wireless communication in the frequency range of 0.6 GHz to 2 GHz, with a resonant frequency of 1.6 GHz. The primary objective of improving upon the narrow bandwidth and low gain limitations of conventional rectangular microstrip patch antennas was achieved through the use of the Z-shaped patch geometry.

The Z-shaped modification alters the surface current distribution across the radiating patch, creating an effective broadband response. Simulation results in Ansys HFSS version 14 confirmed excellent impedance matching, with a return loss of -48 dB and a VSWR of approximately 1.1 at the resonant frequency — both significantly better than the minimum acceptable thresholds for practical antenna design. The antenna demonstrated a stable radiation pattern at 1.6 GHz across both H-plane and E-plane configurations.

The design uses FR-4 glass epoxy substrate, making it cost-effective, widely reproducible, and compatible with standard PCB fabrication processes. The compact dimensions (patch: 56 mm × 50 mm; ground plane: 100 mm × 75 mm) allow it to be integrated into modern handheld and portable communication devices with minimal impact on overall device form factor.

Despite certain limitations such as moderate gain and substrate dielectric losses, the antenna's strong broadband performance, excellent impedance matching, and compact size make it highly suitable for real-world applications in wireless communication, GPS, L-band satellite systems, and mobile networks.

Future work could explore the use of advanced low-loss substrate materials, array configurations for improved directivity, and integration with multiband feed networks to further enhance performance for emerging 5G and IoT applications.

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